Benudhar Behera

Lead Software Engineer | Java, Distributed Systems, Cloud & AI Platforms

Bangalore, India | +91- 9938636246 | benudharbehera003@gmail.com | linkedin.com/in/benudhar | github.com/benudhar

PROFESSIONAL SUMMARY

Lead Backend Engineer with **9 years** of experience architecting and delivering scalable, resilient distributed systems and cloud-native microservices. Deep expertise in Java, Spring Boot, Kafka, Python, FastAPI and AWS, with hands-on experience building GenAI platforms using **RAG, LLMs, Vector Databases, and Model Context Protocol (MCP).** Proven track record designing event-driven architectures, high-throughput APIs, and AI-enabled backend platforms with a focus on scalability, reliability, and clean architecture.

TECHNICAL SKILLS

Programming Languages: Java (Primary), Python

Backend & Frameworks: Spring Boot, Spring WebFlux, FastAPI, REST APIs, Microservices

AI & GenAI: RAG, LLMs, OpenAI API, Spring AI, Model Context Protocol (MCP), AI Agents, Vector Databases, LangChain

Distributed Systems & Messaging: Kafka, Event-Driven Architecture, SQS, SNS

Database & Storage: PostgreSQL, MongoDB, DynamoDB, Redis, Elasticsearch

Cloud & DevOps: AWS, Terraform, Docker, Kubernetes, Jenkins, CI/CD

Architecture: Distributed Systems, Microservices Architecture, Cloud-Native Architecture, System Design, Resilience

Design Principles: System Design, HLD, LLD, Scalability, Design Patterns, SOLID, Rate Limiting

Process & Delivery: Agile, Scrum, Sprint Planning, Stakeholder Management, Cross-functional Collaboration

KEY ACHIEVEMENTS

- **GenAI Platform Impact – EY**: Sustained 600+ requests/minute at low latency for an enterprise GenAI platform supporting document retrieval and AI tool integration.
- **Performance Turnaround – Wipro:** Improved API response times by **40%+** through strategic caching, JVM tuning, and database query optimization on a large-scale enterprise platform.
- **Reliability at Scale – ITC InfoTech:** Engineered **idempotent and failure-resilient payment workflows**, ensuring reliable transaction processing across a live e-commerce order pipeline.
- **Enterprise Application Development – Silicon Tech Lab:** Developed and delivered ERP automation modules that reduced manual business-process effort from day one of my career.

PROFESSIONAL EXPERIENCE

Lead Backend Engineer (Senior Technology Consultant) | Ernst & Young (EY)

Bangalore, India • Mar 2024 - Present

- Architected and delivered a production-grade GenAI conversational platform integrating LLMs with large-scale research datasets, supporting **600+ requests/minute with low-latency**, secure, and highly available document retrieval.
- Engineered cloud-native microservices using **Java 21, Spring Boot, and Kafka** following Hexagonal Architecture, enabling scalable and maintainable enterprise applications.
- Built **RAG-based knowledge retrieval systems** using vector embeddings and semantic search, improving contextual response accuracy and retrieval efficiency across large-scale datasets.
- Designed **MCP-based AI tool integrations** enabling LLMs to dynamically discover and invoke backend capabilities, connecting conversational AI workflows with enterprise services and data.
- Reduced API latency by **~40%** through caching, query optimization, JVM tuning, and asynchronous processing, improving system throughput and responsiveness.
- Partnered with product managers, AI engineers, and data teams to deliver secure, production-ready AI solutions using Agile delivery practices.

Senior Backend Engineer (Senior Project Engineer) | Wipro Limited

Bangalore, India • Jul 2021 – Feb 2024

- Built cloud-native microservices using **Java, Spring Boot, and Kafka** for a large-scale enterprise platform, enabling scalable backend services and real-time event processing.
- Improved API response time by **over 40%** through query optimization, caching strategies, asynchronous processing, and JVM tuning.
- Optimized database indexing strategies and SQL execution plans to improve throughput for high-volume transactional workloads.
- Developed comprehensive unit and integration test suites using **JUnit 5 and Mockito**, improving application stability and reducing production defects.
- Resolved production incidents using **Splunk, Grafana, and AWS CloudWatch**, reducing mean time to resolution through proactive monitoring and log analysis.

Backend Engineer (Associate IT Consultant) | ITC InfoTech

Bangalore, India • Sep 2020 – Jul 2021

- Developed scalable cloud-native backend services using **Java, Spring Boot, and RESTful APIs** for an enterprise e-commerce platform, supporting order lifecycle management, payment processing, and real-time transaction workflows.
- Implemented event-driven architecture using **Kafka** to support asynchronous order processing, payment notifications, and inventory synchronization across distributed services.
- Designed idempotent APIs and resilient payment workflows using retry and failure recovery mechanisms, ensuring reliable transaction processing.
- Built reusable business components following **SOLID principles** and clean architecture, improving code quality and maintainability.
- Increased test coverage and production stability through comprehensive automated testing using JUnit 5 and Mockito.

Software Engineer | Silicon Tech Lab Pvt Ltd.

Bhubaneswar, India • Nov 2017 – Sep 2020

- Developed backend modules using **Java, Spring MVC, Hibernate ORM, and PostgreSQL** for enterprise ERP applications supporting business process automation.
- Implemented business logic and reusable service components supporting core ERP functionalities, improving maintainability and reducing development effort.
- Resolved production issues through root cause analysis, bug fixing, and performance tuning, improving application stability and customer satisfaction.
- Collaborated with senior engineers, QA teams, and business analysts to deliver production-ready features following Agile development practices.

CERTIFICATIONS

- EY IT Architecture - Solution Architecture (Bronze), 2025
- EY Artificial Intelligence - Applied AI (Bronze), 2024
- Introduction to Generative AI - Google Cloud

EDUCATIONAL QUALIFICATION

Bachelor of Technology (B. Tech) — Computer Science & Engineering
Silicon Institute of Technology, Bhubaneswar | 2013 – 2017